

DEVOPS SESSION FOUR





Welcome

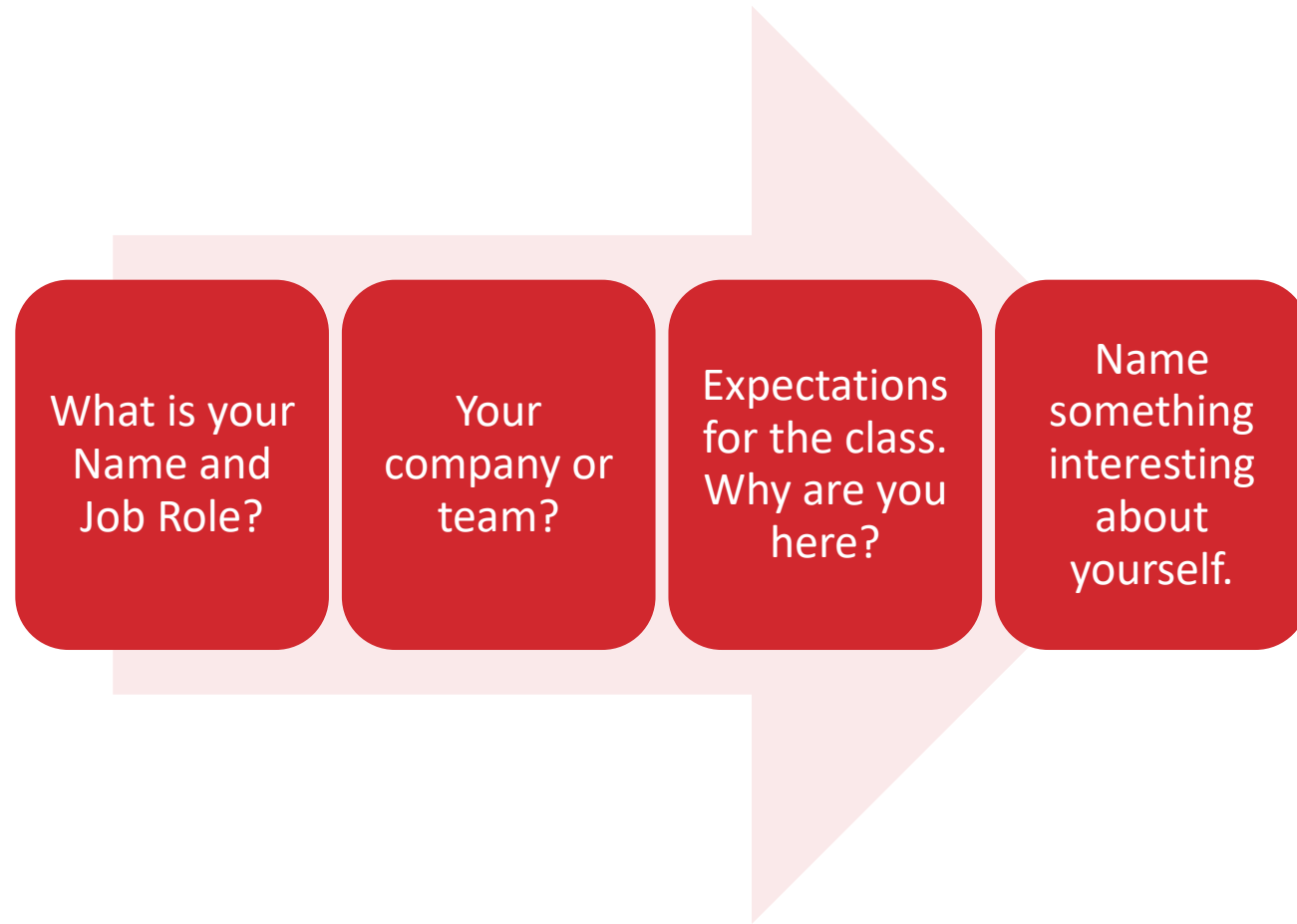
Logistics (breaks, facilities, lunch, etc.)

Rules of Engagement

Introductions

Lets Get Started!

Introductions



Workshop Goals and Structure

- Four-Hour Sessions
- Key DevOps Principles
- Blend of Practical Insights and Demos



What to expect from this workshop

- Flexibility
- Conversations
- Literacy and awareness on the many principles, tools and practices associated with this thing called “DevOps”
- A priority of focus on human behavior first, technology and tools second
- A lot of talk about organizational culture
- An effort to focus on your own situations and challenges so you can act on what you learn



What not to expect from this workshop

- Prescriptions and formulas, rigid processes, step-by-step instructions
- Big overnight transformations
- Perfect solutions that work for everyone
- Extended technical discussions or deep focus on any specific engineering tool



Presenter Information

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- Consultant - Product Development Leader - Agile Coach - Scrum.org Professional Trainer - Mechanical & Aerospace Design Engineer - Analyst
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DEVOPS SESSION 4

DevOps Session 4

Session 4 Agenda

- Recap
- Leadership in DevOps Transformation
- Scaling Strategies
- Refining Metrics for Enterprise Scale
- Integrating Security into DevOps
- Case Study on Enterprise DevOps Success
- Quiz
- Closing & Next Steps



❖ Recap

- Kahoot Quiz on Session 3
- Session 3 Strategy Recap
- Session 3 Readiness Recap
- Session 3 Metrics Recap
- Session 3 Architecture Recap
- Key Takeaways Recap



Kahoot Quiz on Session 3



- Interactive 5-question quiz
- Covers strategy & metrics
- Uses Kahoot platform
- Results guide review
- Sets energetic tone

Session 3 Strategy Recap

How strategy shaped last Session's discussions.

- Aligning DevOps with business goals
- Setting SMART objectives
- Phased rollouts for broad impact
- Linking DevOps to revenue



Session 3 Readiness Recap



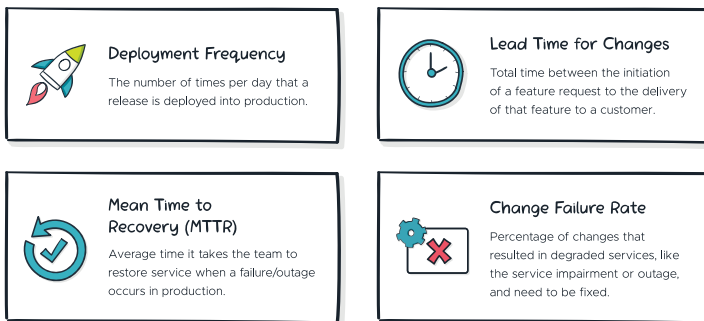
Assessing readiness from last session.

- Culture & tech readiness
- Pain points & constraints
- Tools like DORA, CALMS
- Avoiding chaos

Session 3 Metrics Recap



Key DORA Metrics



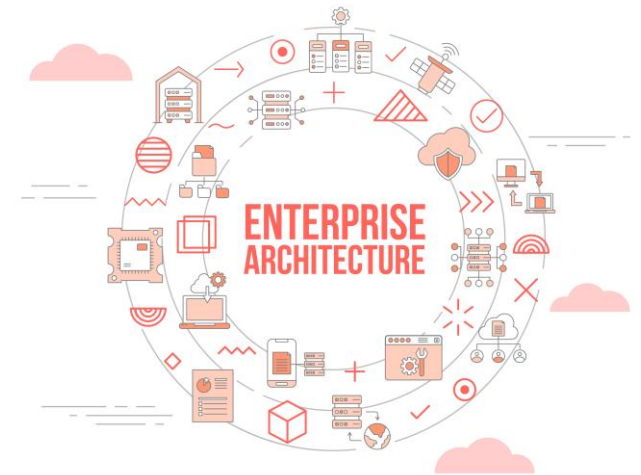
Data driving progress.

- DORA's four metrics
- Time to restore, deployment frequency
- Going beyond basics
- Continuous improvement

Session 3 Architecture Recap

Technology underpinning last Session's progress.

- Pipelines & CI/CD
- Cloud or hybrid setups
- Containerization & microservices
- Observability frameworks



Key Takeaways Recap



Core lessons from Session 3.

- Strategy aligns DevOps with ROI
- Readiness check prevents failure
- DORA metrics track progress
- Architecture fosters speed
- Foundation for enterprise scale

❖ Leadership



- Roles
- Vision Setting Details
- Resource Allocation Tips
- Building Buy-In
- Overcoming Resistance
- Driving Culture Change

Roles

Your critical role in DevOps success.

- Remove silos
- Champion change
- Monitor progress



Vision Setting Details



Crafting a DevOps vision that sticks.

- Define clear goals
- Link to business wins
- Inspire team action
- Share widely & often

Resource Allocation Tips

Smart ways to fund DevOps scale.

- Prioritize key teams
- Budget for tools
- Hire or train staff
- Plan for growth



Building Buy-In



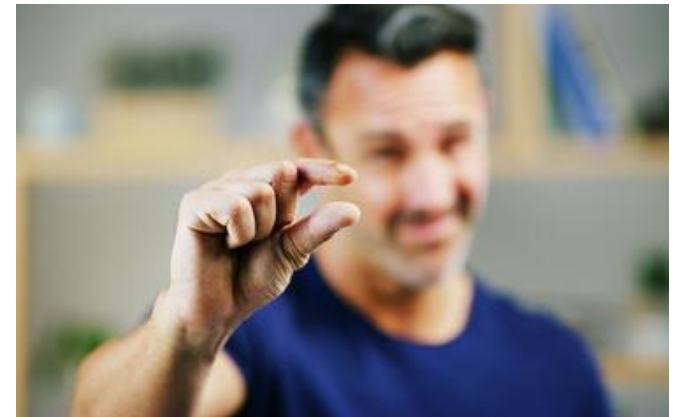
Gaining support for DevOps transformation.

- Communicate benefits
- Show quick wins
- Address resistance
- Engage stakeholders
- Use success stories

Overcoming Resistance

Tackling pushback to DevOps change.

- Listen to concerns
- Show data proof
- Start with small scale
- Highlight benefits



Driving Culture Change



Shifting to a DevOps mindset.

- Foster collaboration
- Encourage experimentation
- Reduce blame
- Reward innovation
- Model behavior

❖ Scaling Strategies

- Scaling Frameworks Overview
- SAFe in Depth
- LeSS Explained
- Managing Multiple Teams
- Team Sync Tactics
- Avoiding Scaling Mistakes



Scaling Frameworks Overview

- Models for enterprise DevOps rollout.
- SAFe for structure
- LeSS for simplicity
- Spotify model for agility
- Hybrid approaches



SAFe in Depth

How SAFe scales DevOps in large organizations.

- Aligns multiple teams
- Uses PI planning
- Balances agility & structure
- Typically suits big enterprises



LeSS Explained

- Scaling with LeSS simplicity.
- Fewer roles & rules
- Focus on direct collaboration
- Central backlog for multiple teams
- Works for mid-size organizations



Managing Multiple Teams

Coordinating DevOps across squads.

- Standardize key processes
- Share best practices
- Align goals and timelines
- Use consistent tooling
- Track multi-team progress



Team Sync Tactics



Keeping teams aligned at scale.

- Regular cross-team meetings
- Shared dashboards & boards
- Cross-training or skill rotation
- Unified backlog or roadmaps

Avoiding Scaling Mistakes

Common traps in enterprise DevOps.

- Over-centralization
- Ignoring autonomy
- Scaling too fast
- Neglecting training
- Poor communication





❖ Refining Metrics for Enterprise Scale

- Beyond DORA Basics
- Business Outcome Metrics
- Enterprise Metric Examples
- Team Health Metrics
- Dashboard Refinement

Beyond DORA Basics

- Expanding metrics for enterprise scale.
- Add business outcomes
- Track adoption rates
- Assess cost efficiency
- Link to strategic goals



Business Outcome Metrics



Metrics tied to real business impact.

- Revenue growth
- Customer retention
- Market share
- Time-to-market

Enterprise Metric Examples

- Specific metrics for large orgs.
- Deployment cost
- Team velocity
- Customer satisfaction
- Security incidents
- Process efficiency



Team Health Metrics

- Gauging team strength at scale.
- Morale surveys
- Turnover rates
- Burnout signals
- Collaboration scores



Dashboard Refinement

Enhancing metrics visibility.

- Aggregate data
- Highlight trends
- Customize for roles
- Automate updates
- Add alerts



❖ Integrating Security

- DevSecOps Basics
- Shift Left Explained
- Security Practices
- Compliance in DevOps
- Tools for Security



DevSecOps Basics

Introduction to security in DevOps.

- Shift security left
- Integrate early & often
- Automate checks
- Balance speed & safety
- Team ownership



Shift Left Explained



Catching security early in DevOps.

- Fix in dev phase
- Reduce production risks
- Speed up fixes
- Lower overall costs

Security Practices

- Key methods for secure DevOps.
- Code reviews
- Vulnerability scans
- Compliance checks
- Threat modeling
- Access controls



COMPLIANCE



Compliance in DevOps

- Meeting rules in DevOps scale.
- Align with regulations
- Automate audit trails
- Track compliance continuously
- Train teams on standards

Tools for Security

- Essential security tools for DevOps.
- Snyk for code scanning
- Azure Security Center
- Dependabot
- Static analysis tools
- Runtime protection



Case Study on Enterprise DevOps Success

- Industry Example
- Challenges Faced
- Legacy Tech Issues
- Solutions Applied
- Automation Details
- Results Achieved
- Lessons Learned





Industry Example

- A large enterprise adopting DevOps.
- Financial services scope
- Multiple teams, global
- Heavy regulations
- Customer-facing apps

Challenges Faced



Obstacles in scaling DevOps.

- Siloed teams
- Sluggish release cycles
- Security compliance
- Cultural inertia
- Budget constraints



Legacy Tech Issues

Old systems slowing DevOps scale.

- Mainframe dependencies
- Monolithic codebase
- Manual configuration
- Skill gaps



Solutions Applied

How they overcame obstacles.

- Adopted a scaled framework
- Introduced Infrastructure as Code
- Implemented trunk-based dev
- Shifted security left
- Provided extensive training



Automation Details

- How automation powered their success.
- CI/CD pipelines for all apps
- Automated testing from unit to integration
- Self-service provisioning for squads
- Observability with real-time dashboards



Results Achieved

- Outcomes of their DevOps transformation.
- 40% cut in release cycle time
- 25% drop in incidents
- Zero major security breaches for 2 years
- Increased developer satisfaction



Lessons Learned

Takeaways from their DevOps journey.

- Start with a strong pilot
- Invest in training & culture
- Integrate security from day one
- Use data to drive decisions
- Celebrate wins, no matter the size



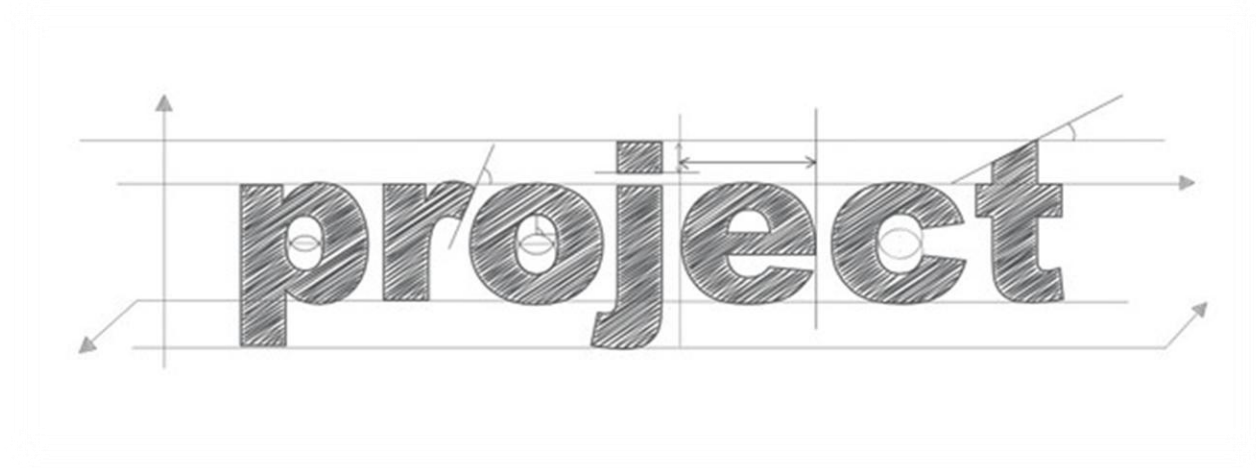
Lab Overview

- Purpose
- What You'll Accomplish
- Estimated Time
- Lab Structure



Step 1: Setting Up Azure DevOps Project

- Sign in or create Microsoft account
- Access dev.azure.com
- Create a new project
- Choose project name & process



Step 2: Creating a Basic Repository

- Go to Repos in new project
- Create or import a repo
- Add a README file
- Copy-paste sample content



Step 3: Creating the Pipeline

- Go to Pipelines → New
- Select your repo
- Use a starter YAML
- Echo a message in the pipeline
- Save & run



Save

Step 4: Testing and Observing Pipeline Output

- View pipeline logs
- Confirm success status
- Optional: Add more steps
- Rerun pipeline



Step 5: Setting Up a Dashboard

- Go to Dashboards
- Create a new dashboard
- Add pipeline widget
- (Optional) add queries or charts
- Observe real-time data



Group Reflection

- Biggest takeaway
- Potential uses in your org
- Security or compliance integration
- Next steps



❖ Quiz



- Which leadership action is crucial for scaling DevOps?
- Which framework is known for adding structure to large DevOps rollouts?
- Which metrics best suit an enterprise looking beyond DORA?
- What is the main advantage of shifting security left?
- Which result did the case study enterprise achieve?

Question 1

Which leadership action is crucial for scaling DevOps?

- A. Micromanage daily tasks
- B. Providing a unifying vision
- C. Avoiding resource commitments
- D. Delegating all decisions to middle management



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Which leadership action is crucial for scaling DevOps?

- A. Micromanage daily tasks
- B. **Providing a unifying vision**
- C. Avoiding resource commitments
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Question 2

Which framework is known for adding structure to large DevOps rollouts?

- A. LeSS
- B. Spotify
- C. SAFe
- D. None of the above



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Which framework is known for adding structure to large DevOps rollouts?

- A. LeSS
- B. Spotify
- C. **SAFe**
- D. None of the above



Question 3

(Choose 2) Which metrics best suit an enterprise looking beyond DORA?

- A. Team velocity
- B. Lead time
- C. Customer satisfaction
- D. MTTR



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Question 4

What is the main advantage of shifting security left?

- A. Catching vulnerabilities early
- B. Slowing releases for thorough tests
- C. Hiding compliance checks
- D. Delaying fixes until production



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Question 5

Which result did the case study enterprise achieve?

- A. Increased major breaches
- B. Slower time to market
- C. Reduced incidents by 25%
- D. Stagnant developer morale



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Closing & Next Steps



Summarizing Session 4's lessons and preparing for the upcoming session.

- Final Thoughts on Leadership
- Action Items: Scaling Tactics
- Upcoming Topics: Hybrid Cloud, Continuous Testing
- Thank You